

GENERALIZATION OF A LEMMA OF G. F. ROSE*

I. L. GÁL, J. B. ROSSER and D. SCOTT

In attempting to reconstruct Rose's proof of Lemma 3.2 of [1], the present authors found what is apparently a different and simpler method, which moreover leads to a far stronger conclusion.

We are operating in the Heyting propositional calculus as formulated on p. 3 of [1] or on pp. 82 and 101 of [2], and shall make use of relevant theorems on pp. 90, 113-119 of [2]. We shall use a, b, c, w, x, y, z as propositional variables.

We say that a conjunction is simple if each factor has one of the forms: (i) a , (ii) $\neg a$, (iii) $a \supset b$, (iv) $a \supset (b \vee c)$, (v) $(a \& b) \supset c$, (vi) $(a \supset b) \supset c$.

THEOREM. *Each formula $B \supset C$ is interdeducible with some formula $A \supset C$, where A is a simple conjunction.*

PROOF by induction on the number of symbols in B . If B has one or two symbols, it must have the form (i) or (ii) and the theorem holds. So let B have three or more symbols, and assume the theorem holds for all cases where B has fewer symbols.

CASE 1: B has a subformula of the form $\neg x$. Let w, z be variables new to B and C , and let B' be the result of replacing $\neg x$ by z in B . Then $B \supset C$ is interdeducible with

$$(1) \quad B' \supset (D \supset C),$$

where D is $[((x \supset z) \supset z) \& \neg w \& (z \& x \supset w)]$. For we have $\vdash \neg x \supset (x \supset z)$, and hence $(x \supset z) \supset z \vdash \neg x \supset z$. Also $(z \& x) \supset w \vdash z \supset (x \supset w) \vdash z \supset (\neg w \supset x) \vdash \neg w \supset (z \supset x)$, so that $(z \& x) \supset w, \neg w \vdash z \supset x$. Therefore $D \vdash z \sim \neg x$, and so by the replacement theorem (1) follows from $B \supset C$. Conversely, $B \supset C$ follows from (1) by replacing z by $\neg x$ and w by $\neg x \& x$. Since B' contains fewer symbols than B , there is a simple conjunction A' such that (1) is interdeducible with $A' \supset (D \supset C)$, and so with $A \supset C$, where A is the simple conjunction $A' \& D$.

CASE 2: B has a subformula of the form $x \supset y$. Take B' to be the result of replacing $(x \supset y)$ by z in B , let D be $[((x \supset y) \supset z) \& ((z \& x) \supset y)]$, and proceed as in Case 1.

CASES 3 and 4: B has a subformula of the form $x \& y$ or $x \vee y$. Take B' to be the result of replacing $x \& y$ or $x \vee y$ by z in B , and let D be $[((x \& y) \supset z) \& (z \supset x) \& (z \supset y)]$ or $[(x \supset z) \& (y \supset z) \& (z \supset (x \vee y))]$, respectively.

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CORNELL UNIVERSITY and PRINCETON UNIVERSITY